

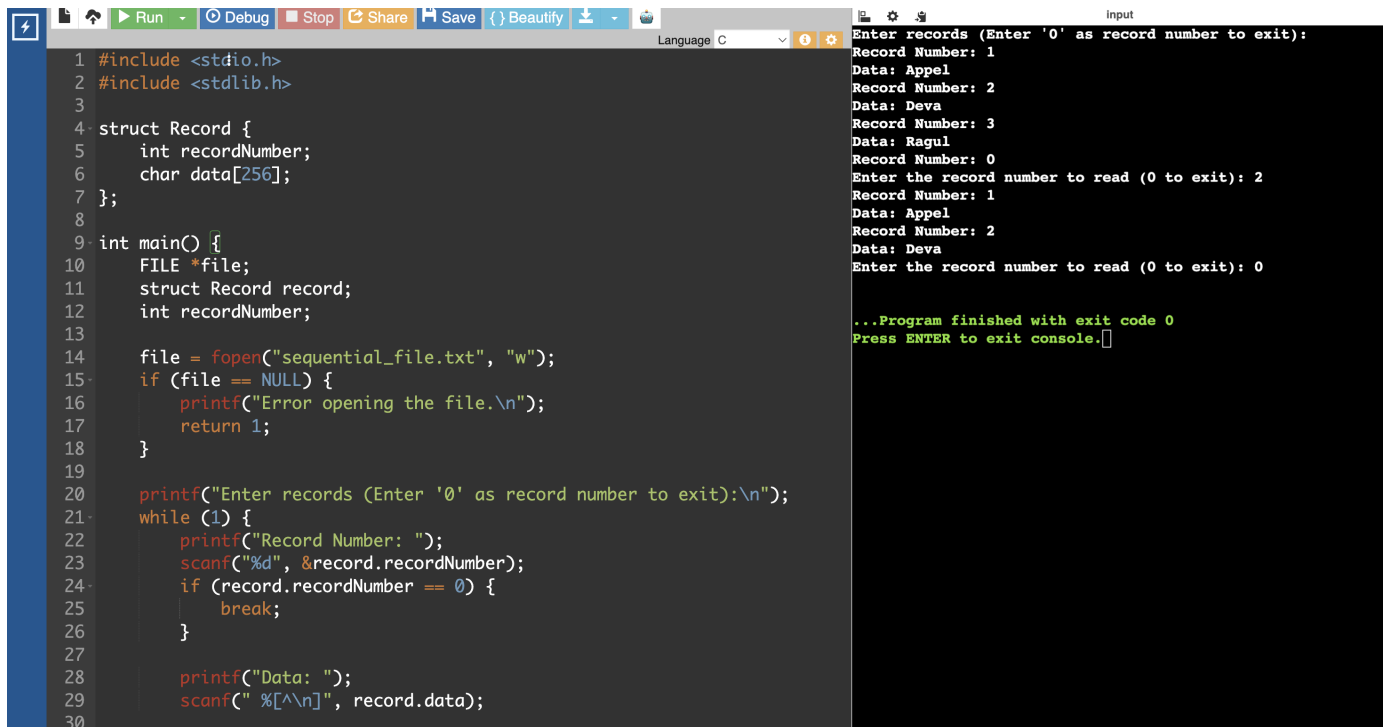
CSA0404 – OPERATING SYSTEMS

EXPERIMENT 34

QUESTION

Consider a file system where the records of the file are stored one after another both physically and logically. A record of the file can only be accessed by reading all the previous records. Design a C program to simulate the file allocation strategy.

PROGRAM



```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Record {
5     int recordNumber;
6     char data[256];
7 };
8
9 int main() {
10     FILE *file;
11     struct Record record;
12     int recordNumber;
13
14     file = fopen("sequential_file.txt", "w");
15     if (file == NULL) {
16         printf("Error opening the file.\n");
17         return 1;
18     }
19
20     printf("Enter records (Enter '0' as record number to exit):\n");
21     while (1) {
22         printf("Record Number: ");
23         scanf("%d", &record.recordNumber);
24         if (record.recordNumber == 0) {
25             break;
26         }
27
28         printf("Data: ");
29         scanf("%[^\n]", record.data);
30     }
```

Input

```
Enter records (Enter '0' as record number to exit):
Record Number: 1
Data: Appel
Record Number: 2
Data: Deva
Record Number: 3
Data: Ragul
Record Number: 0
Enter the record number to read (0 to exit): 2
Record Number: 1
Data: Appel
Record Number: 2
Data: Deva
Enter the record number to read (0 to exit): 0

...Program finished with exit code 0
Press ENTER to exit console.
```